

# Qianyi Liu

Ph.D. Student, Department of Civil Engineering, Stony Brook University  
qianyi.liu@stonybrook.edu | Stony Brook, NY, USA | ASCE Student Member | AGU Student Member

## EDUCATION

### Stony Brook University

*Ph.D. in Civil Engineering; GPA: 4.00/4.00*

- Research focus: geotechnical/geohazard engineering, landslide susceptibility mapping, GIS, spatial machine learning, causal and explainable AI. Advisor: Prof. Te Pei.

08/2025–Present

*Stony Brook, NY, USA*

### University of Macau

*M.S. in Coastal Environment and Safety; GPA: 3.25/4.00*

- Relevant modules: Hydrodynamics and Floods (A); Advanced Computational Methods: Principles and Applications (A-).

08/2023–06/2025

*Macau, China*

### Capital University of Economics and Business

*B.Eng. in Safety Engineering; GPA: 3.81/5.00 (88.2/100)*

- Relevant modules: Electrical Safety Engineering (100), Systems Engineering (100), Engineering Drawing II (99), Python Data Analysis (93).

09/2019–06/2023

*Beijing, China*

## RESEARCH INTERESTS

Geotechnical engineering; landslide susceptibility mapping; geohazard assessment; trustworthy AI; causal and explainable machine learning; spatially aware machine learning; GIS and remote sensing; hydrodynamic and compound-flood modeling; spatial cross-validation; SHAP analysis; infrastructure resilience.

## RESEARCH EXPERIENCE

### Landslide Susceptibility Mapping and Geohazard Assessment

*Ph.D. Researcher, Stony Brook University*

- Developing geospatial machine-learning workflows for landslide susceptibility and slope-related geohazard assessment, emphasizing physically meaningful, spatially robust, and interpretable outputs.
- Integrating terrain, hydrological, soil-related, vegetation, and rainfall-related factors to analyze spatially varying slope-instability mechanisms.
- Applying spatial cross-validation, explainable AI, causal-graph analysis, and GIS-based mapping to improve model reliability and engineering interpretability.

08/2025–Present

*Stony Brook, NY, USA*

### Flood Susceptibility Mapping with Explainable Ensemble Machine Learning

*Research Assistant, University of Macau*

- Conducted research on flood susceptibility mapping using GIS, multi-source environmental data, hydrodynamic modeling, and explainable ensemble machine-learning methods.
- Developed flood susceptibility maps for the Greater Bay Area / Pearl River Delta case study and analyzed feature contributions using SHAP-based interpretation.

08/2023–08/2024

*Macau, China*

### Smart Emergency Response Development Strategy (2035)

*National Key Project; Team Member; Lead: Academician Weicheng Fan*

- Synthesized national macro policies related to smart emergency management and contributed to drafting the policy environment and national demand section for smart emergency development.

09/2020–10/2020

*China*

## PUBLICATIONS AND MANUSCRIPTS

### Peer-Reviewed Journal Article

- Ma, S., Liu, Q., & Zhang, Y. (2021). A prediction method of fire frequency: Based on the optimization of the SARIMA model. *PLOS ONE*, 16(8), Article e0255857. <https://doi.org/10.1371/journal.pone.0255857>.

### Manuscripts Under Review / Decision in Process

- Du, H., Fei, K., Liu, Q., & Zhang, L. (2026). *Tropical Cyclone-Induced Compound Flooding Reveals Critical Flood Zone Transitions in the Pearl River Delta*. Submitted to *Ocean Engineering*. Status: Under Review.
- Liu, Q., Gao, L., Du, H., Pei, T., Kuok, S.-C., & Chen, S. (2026). *Integrating Explainable Ensemble Machine Learning with Hydrodynamic Modeling to Assess Tropical Cyclone-Induced Flood Susceptibility*. Submitted to *Geoscience Frontiers*. Status: Decision in Process.

## CONFERENCE ABSTRACTS AND PRESENTATIONS

- Liu, Q., Liu, X., Pei, T., Devineni, N., & Tian, Y. (2025). Understanding Geo-Environmental Factors for Landslide Occurrences Using Geographically Explainable AI: A Case Study from the Northeastern United States. *AGU Fall Meeting Abstracts*, 2025, NH33F-0502.
- Liu, Q., Liu, X., & Pei, T. (2026). Trustworthy AI for Geohazard Mitigation: A Causal and Explainable Framework for Landslide Susceptibility Mapping. 2026 NSF NHERI GSC Mini-Conference.
- Liu, Q., Liu, X., & Pei, T. (2026). Trustworthy AI for Geohazard Mitigation: A Causal and Explainable Framework for Landslide Susceptibility Mapping. SUNY AI Symposium.
- Liu, Q., Liu, X., Pei, T., Devineni, N., & Tian, Y. (2027). Physics-Guided and Spatially Aware Machine Learning for Trustworthy Landslide Susceptibility Mapping in Heterogeneous Terrain. ASCE 2027 Abstract.
- Liu, Q., Liu, X., Pei, T., Cui, H., Devineni, N., & Tian, Y. (2027). Causality-Constrained Explainable Machine Learning for Identifying Spatially Varying Controls in Landslide Susceptibility Mapping. ASCE 2027 Abstract.

## TEACHING EXPERIENCE

---

### Department of Civil Engineering, Stony Brook University

Teaching Assistant

08/2025–Present

Stony Brook, NY, USA

- Support course instruction, grading, laboratory/problem-solving sessions, and student guidance for undergraduate civil engineering courses.
- Help students connect experimental or computational results with civil engineering concepts and reporting standards.

## ACADEMIC SERVICE

---

### 2026 NSF NHERI GSC Mini-Conference

04/2026

Reviewer, NSF NHERI Graduate Student Council

- Reviewed student conference submissions and contributed to the academic review process for the 2026 NHERI GSC Mini-Conference.

## ADDITIONAL PROFESSIONAL EXPERIENCE

---

### China Aerospace Construction Group Co., Ltd.

06/2022–07/2022; 09/2022–10/2022

Intern

Beijing, China

- Compiled safety inspection records for the Pinggu Men Station natural gas project, organized ledgers, and developed educational safety materials and emergency response plans.
- Supported implementation of corporate safety protocols and enhanced execution methods for safety management under pandemic-related constraints.

### Beijing GuoXinAn Technology Co., Ltd.

07/2021–08/2021

Intern

Beijing, China

- Assisted SMEs with risk-source declaration in Changping District through direct offline engagement and online reporting support.
- Supported community-level enterprise safety and regional risk management practices.

## AWARDS AND HONORS

---

- Second Prize, National College Students' Mathematics Modelling Competition, Team Leader, 09/2021.
- Second Prize, National College Students' Mathematics Modelling Competition, 09/2020.
- Honorable Mention, Mathematical Contest in Modeling (MCM/ICM), 09/2021.
- Second Prize, Beijing Energy Conservation and Emission Reduction Competition, Team Leader, 09/2021.
- Key Project Leader, Summer Social Practice by the College, 09/2021.

## PROFESSIONAL MEMBERSHIPS

---

- American Society of Civil Engineers (ASCE), Student Member.
- American Geophysical Union (AGU), Student Member.

## TECHNICAL SKILLS

---

- **Geospatial analysis:** ArcGIS Pro, ArcMap, GIS-based spatial analysis, raster data processing, susceptibility mapping.
- **Programming and data analysis:** Python, Jupyter Notebook, MATLAB, statistical analysis, Microsoft Office Suite.
- **Machine learning and interpretation:** Ensemble machine learning, trustworthy AI, causal and explainable AI, SHAP analysis, model evaluation and visualization.
- **Specialized software:** SPSS, Stata, Citespace, Origin. **Languages:** Mandarin Chinese (native), English (fluent). **Certificates:** CET-4, CET-6.

## SELECTED INTERESTS

---

Swimming; model aviation; Pokemon Trading Card Game.